

MLPR - Lab 8

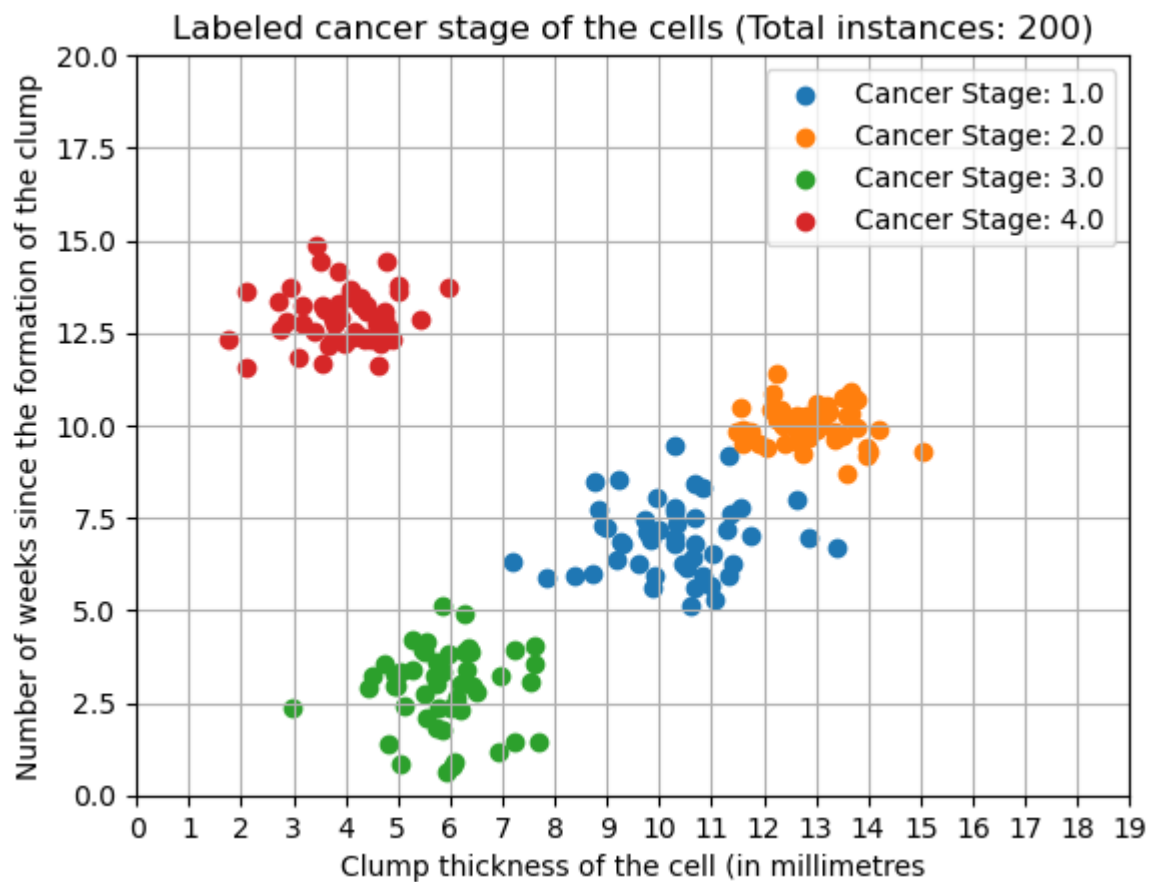
INSTRUCTIONS

Step 1: Import all libraries

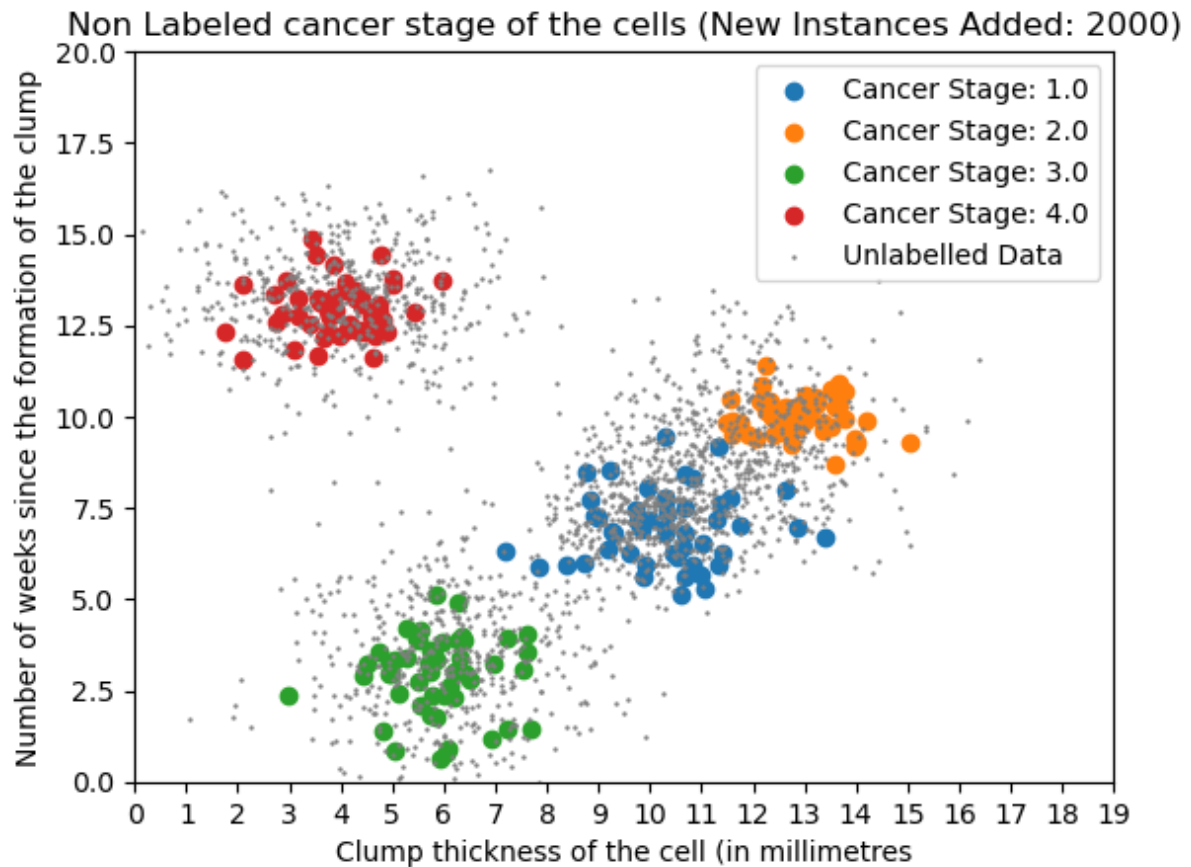
- Numpy
- Pandas
- Pyplot
- Self Training classifier from sklearn
- K Nearest neighbour classifier
- Confusion matrix, classification report, accuracy score, balanced accuracy score

Step 2: Read the data given in data.csv

Step 3: show a scatter plot of 'Clump thickness vs No of week' and plot their class labels given in column 'Cancer stage' (1, 2, 3, 4) for all 200 datapoints. See the reference output image below.

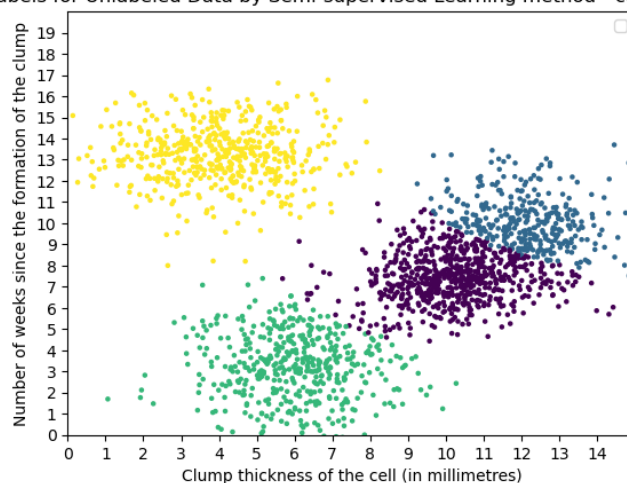


Step 4: Now, Plot the other added 2000 datapoints given in columns 'Clump thickness_new' and 'No of week_new' over the previous scatter plot without using their class label. See the reference output image below.

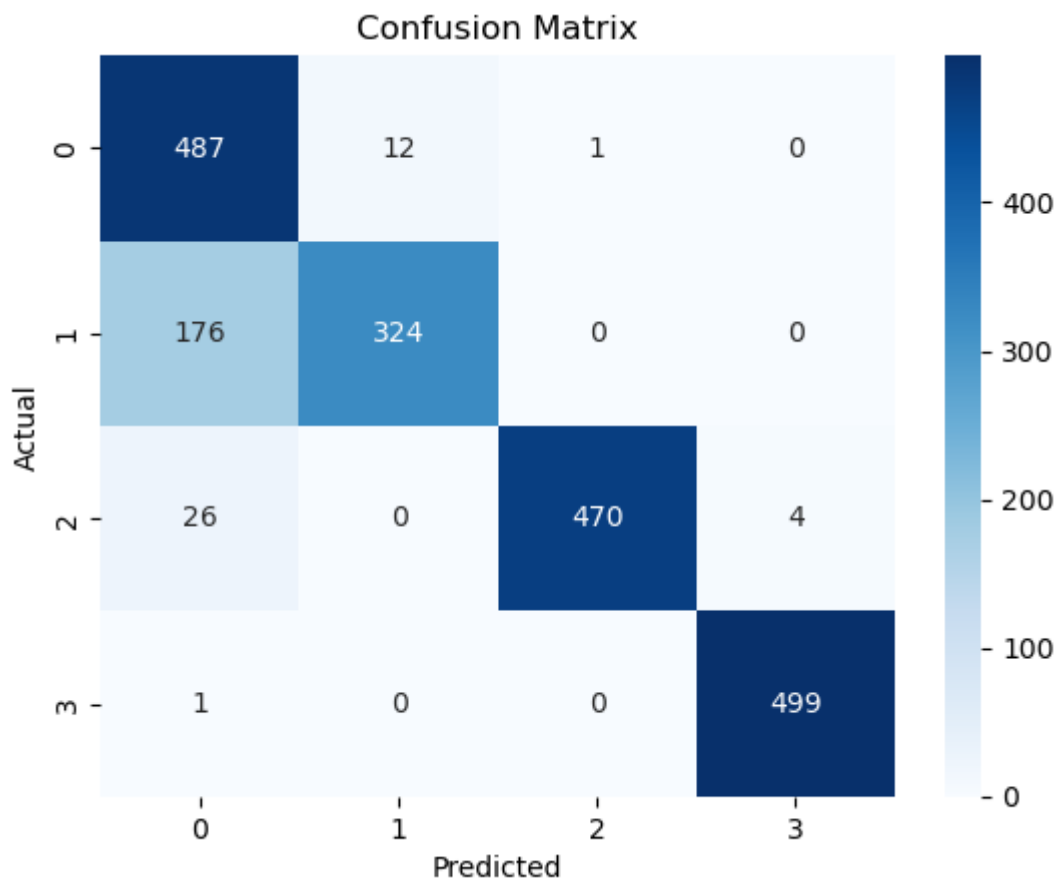


Step 5: Now train the semi supervised model for the labeled 200 datapoints and make predictions for class label of new 2000 added datapoints and plot the scatters. Use KNN for base estimator. Use `Model.fit()` and `model.predict()` for training and prediction.

Fitted Labels for Unlabeled Data by Semi-supervised Learning method - cell cancer stage



Step 6: Now your added 2000 new data have their predicted class and true class are given in column 'True cancer stage'. Compute and print accuracy score, plot classification report and the confusion matrix.



Report:

Answer the following 5 questions within your report:

1. Provide three applications of Semi-supervised learning.
2. What are the three assumptions of Semi-supervised learning?
3. What is significance each of using the above three assumptions in Semi-supervised Learning?
4. How does the Co-training method differ from the Self-training method of Semi-supervised learning? Which one usually performs better for accuracy and such performance metrics?
5. How to evaluate the performance of a semi-supervised learning method while training?

DELIVERABLES: Items to be uploaded on LMS-

- Main code file along with the outputs and the report answers in one file in **PDF format**. Ensure that your file contains the scatter plots for all the output images above, Confusion matrix etc.
- **Please note: Write your comments along with the code for each step as required. All outputs and plots must have correct labels, titles, axes titles etc.**
- Report answers must be written in **full and complete sentences**. Answer each question in a new line.
- Your file should be as follows. **Yourname_lab8.pdf**
- Due time and date are given on LMS. Submit it before the deadline.